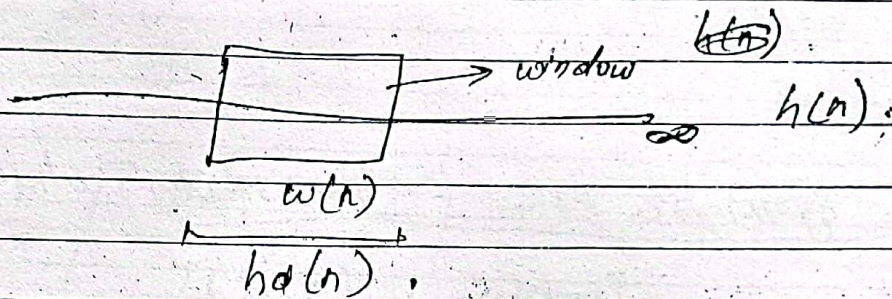


6

FIR Filter Design

Name $w(n)$

Rectangular

1

Triangular (Bartlett)

$$1 - \frac{2|n - \frac{M-1}{2}|}{M-1}$$

 $M-1$

Blackman

$$0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.8 \cos \frac{4\pi n}{M-1}$$

Hamming

$$0.54 - 0.46 \cos \frac{2\pi n}{M-1}$$

Hanning

$$\frac{1}{2} \left[1 - \cos \frac{2\pi n}{M-1} \right]$$

Concept:

$$H_d(\omega)$$

2 π DTFT

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega.$$

So, $h(n) = h_d(n) \cdot w(n)$

Design symmetric FIR low pass filter, where,

$$H_d(\omega) = e^{-j\omega n} \quad \text{for } |\omega| \leq \omega_c$$

$$= 0, \quad \text{otherwise.}$$

The length of filter must be 7 and $\omega_c = 1$.
Use rectangular window.

e) Soln:

Here, Given, M (order) = 7.
 $\omega_c = 1$.

$$L = \frac{M-1}{2} = \frac{7-1}{2} = 3$$

$$\therefore L = 3$$

Now,

Using DTFT.

$$e^{jx} - e^{-jx} = 2j \sin x.$$

Bad when $n = \tau$,

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$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega \tau} \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\tau)} d\omega \quad [\text{suppose } n \neq \tau.]$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi} \frac{1}{j(n-\tau)} \left[e^{j\omega_c(n-\tau)} - e^{-j\omega_c(n-\tau)} \right]$$

$$= \frac{1}{2\pi j(n-\tau)} \left[2j \sin \omega_c(n-\tau) \right]$$

$$= \frac{\sin \omega_c(n-\tau)}{\pi(n-\tau)}$$

where $n = \tau$.

Then,

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot d\omega$$

$$= \frac{1}{2\pi} [\omega]_{-\omega_c}^{\omega_c}$$

$$= \frac{2\omega_c}{2\pi}$$

$$h_d(n) = \frac{\omega_c}{\pi}$$

$$\therefore h_d(n) = \frac{\sin \omega_c (n - \tau)}{\pi (n - \tau)} ; n \neq \tau$$

$$= \frac{\omega_c}{\pi} ; n = \tau$$

$$h_d(n) = \frac{\sin \omega_c (n - 3)}{\pi (n - 3)} ; n \neq 3$$

$$= \frac{\omega_c}{\pi} ; n = 3$$

Now,

$$h(n) = h_d(n) \cdot W(n)$$

$$h(n) = \frac{\sin \omega_c (n - 3)}{\pi (n - 3)} \cdot 1 ; n \neq 3$$

$$= \frac{\omega_c}{\pi} \cdot 1 ; n = 3$$

$$W(n) = 1, \text{ for } n = 0, \dots, M-1$$

WPN doc

Transition
width

Min. stop band
attn

1. Rectangular

$$\frac{4\pi}{M+1}$$

- 21 dB

2. Bartlett

$$\frac{8\pi}{M}$$

- 25 dB

3. Hanning

$$\frac{8\pi}{M}$$

- 44 dB

4. Hamming

$$\frac{8\pi}{M}$$

- 53 dB

5. Blackman

$$\frac{12\pi}{M}$$

- 74 dB

$w(n)$

1

$$\frac{1 - \left| n - \frac{M-1}{2} \right|}{M-1}$$

$$\frac{1}{2} \left(1 - \frac{\cos 2\pi n}{M-1} \right)$$

$$0.54 - 0.46 \cos \frac{2\pi n}{M-1}$$

$$0.42 - 0.58 \cos 2\pi n$$

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FIR Filter

FIR

Design a LPF digital filter to be used in an A-D - $H(z)$ - D/A structure which will have -3dB at 30π rad/sec and have attⁿ of 50 dB at 45π rad/sec. The filter is required to have linear phase and system uses sampling rate of 100 sam/sec.

⇒ Soln:
Here;

$$\text{Cutoff frequency } (\Omega_c) = 30\pi \text{ rad/sec.}$$

$$\text{Stop band freq. } (\Omega_s) = 45\pi \text{ rad/sec.}$$

$$\text{Sampling Freq } (F_s) = 100 \text{ sam/sec.}$$

$$\text{Stop band att. } (A_s) = \underline{\underline{50 \text{ dB}}}$$

Now, after sampling;

$$\omega_c = \frac{30\pi}{100} = 0.3\pi$$

$$\omega_s = \frac{45\pi}{100} = 0.45\pi$$

Hamming Window

If in question if filter function is not given then by default,

$$H_d(\omega) = e^{-j\omega\tau}, \quad -\omega_c \leq \omega \leq \omega_c$$

$$= 0, \quad \text{otherwise,}$$

Now, applying IDTFT,

$$h(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega.$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\tau} \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\tau)} d\omega$$

when $n \neq \tau$.

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi j(n-\tau)} \left[e^{j\omega_c(n-\tau)} - e^{-j\omega_c(n-\tau)} \right]$$

$$= \frac{1}{2\pi j(n-\tau)} \left[2j \sin \omega_c(n-\tau) \right]$$

$$= \frac{\sin \omega_c(n-\tau)}{\pi(n-\tau)}$$

when $n = \tau$,

$$hd(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega$$

$$= \frac{1}{2\pi} \left[\omega \right]_{-\omega_c}^{\omega_c} = \frac{\omega_c}{\pi}$$

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$$h_d(n) = \frac{8\pi\omega_c(n-\tau)}{\pi(n-\tau)}, \quad n \neq \tau$$

$$h_d(n) = \frac{\omega_c}{\pi}, \quad n = \tau$$

Transition width of main lobe of Hamming window
is $\frac{8\pi}{M}$

General formula of lobe width. = $K\left(\frac{2\pi}{M}\right)$

Now

$$\frac{8\pi}{M} = K \frac{2\pi}{M}$$

$$\therefore K = 4.$$

Now,

order is,

$$N = K \left(\frac{2\pi}{\omega_1 - \omega_2} \right)$$

$$= 4 \left(\frac{2\pi}{0.45\pi - 0.3\pi} \right)$$

$$= 53.33$$

$$N \approx 55$$

$$\therefore T = \frac{55-1}{2}$$

$$T = 27$$

Now,

$$h_d(n) = \frac{\sin 0.3\pi (n-27)}{\pi (n-27)} \quad , \quad n \neq 27$$

$$= 0.3 \quad , \quad n = 27$$

Now,

$$h(n) = h_d(n) \cdot w(n)$$

now here;

$$w(n) = \frac{0.54 - 0.46 \cos \frac{2\pi n}{M-1}}{M-1}$$

$$= \frac{0.54 - 0.46 \cos \frac{2\pi n}{55-1}}{55-1}$$

$$= \frac{0.54 - 0.46 \cos \frac{\pi n}{27}}{27}$$

$$h(n) = h_d(n) \cdot w(n)$$

$$= \frac{\sin 0.3\pi (n-27)}{\pi (n-27)} \times \frac{0.54 - 0.46 \cos \frac{\pi n}{27}}{27} \quad , \quad n \neq 27$$

$$= 0.3 \left[\frac{0.54 - 0.46 \cos \frac{\pi n}{27}}{27} \right] \quad , \quad n = 27$$

$n=27$
 $M=1$

Design a normalized linear phase FIR filter -
having $T=4$ and attenuation of 40dB in stopband.

2) Soln

Here, Given, $T=4$.

Stop band attenuation (A_s) = 40 dB

$$T = \frac{M-1}{2}$$

$$4 = \frac{M-1}{2}$$

82M-

$$\therefore M = 9$$

Hanning window, because ^{min} stop band attenuation given is 40dB.

Now,

$$H_d(\omega) = \begin{cases} e^{-j\omega T} & -\omega_c \leq \omega \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

Now, applying IDTFT

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j\omega T} \cdot e^{j\omega n} d\omega$$

now, continuing the same process, we get

$$h_d(n) = \left. \begin{aligned} &\frac{\sin \omega_c (n-\tau)}{\pi (n-\tau)} & n \neq \tau \\ &\frac{\omega_c}{\pi} & n = \tau \end{aligned} \right\}$$

$\omega_c = 1 \text{ rad/sec}$ [since it is a normalized filter].

$$\therefore h_d(n) = \frac{\sin(n-4)}{\pi(n-4)} \quad , \quad n \neq 4$$

$$= \frac{1}{\pi} \quad , \quad n = 4$$

Now,

$$h(n) = h_d(n) \cdot w(n) \quad \text{--- (i)}$$

for that,

$$w(n) = \frac{1}{2} \left[1 - \cos \frac{n\pi}{4} \right]$$

$$w(n) = \frac{1}{2} \left[1 - \cos \frac{n\pi}{4} \right]$$

putting $w(n)$ in (i)

$$h(n) = \left. \begin{aligned} &\frac{\sin(n-4)}{\pi(n-4)} \cdot \frac{1}{2} \left[1 - \cos \frac{n\pi}{4} \right] & n \neq 4 \end{aligned} \right\}$$

$$= \left. \begin{aligned} &\frac{\sin(n-4)}{\pi} \cdot \frac{1}{2} \left[1 - \cos \frac{n\pi}{4} \right] & n = 4 \end{aligned} \right\}$$

$n=0 \dots M-1$
 $n=0 \dots 8$

Specs $\xrightarrow{\text{B/L/O}} \text{Chebyshev} \xrightarrow{\text{P.E.}} H(z)$

4. A digital LPP is given as

$$H(z) = \frac{0.1 + 0.4z^{-1}}{1 - 0.6z^{-1} + 0.1z^{-2}}$$

with cutoff $\omega_c = 0.2575\pi$. Design high pass filter with $\omega_c = 0.3567\pi$.

2) Soln.

Here,

$$\omega_c = 0.2575\pi$$

$$\omega_c' = 0.3567\pi$$

$$q = \cos\left(\frac{\omega_c - \omega_c'}{2}\right)$$

$$\cos\left(\frac{\omega_c + \omega_c'}{2}\right)$$

* Radial Mode.

$$q = -0.5574$$

Now,

$$H(z) = \frac{0.1 + 0.4 \left[- \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \right]}{1 - 0.6 \left(- \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \right) + 0.1 \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right)}$$

$$= \frac{0.1 + 0.4 \left[- \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \right]}{1 - 0.6 \left(- \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \right) + 0.1 \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right) \left(\frac{z^{-1} - 1}{1 + qz^{-1}} \right)}$$

passband edge = cutoff freq

Kaiser Window:-

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$$W(n) = I_0 \left\{ B \left[1 - \left(\frac{n-\alpha}{\alpha} \right)^2 \right]^{1/2} \right\}$$

$I_0(\beta)$

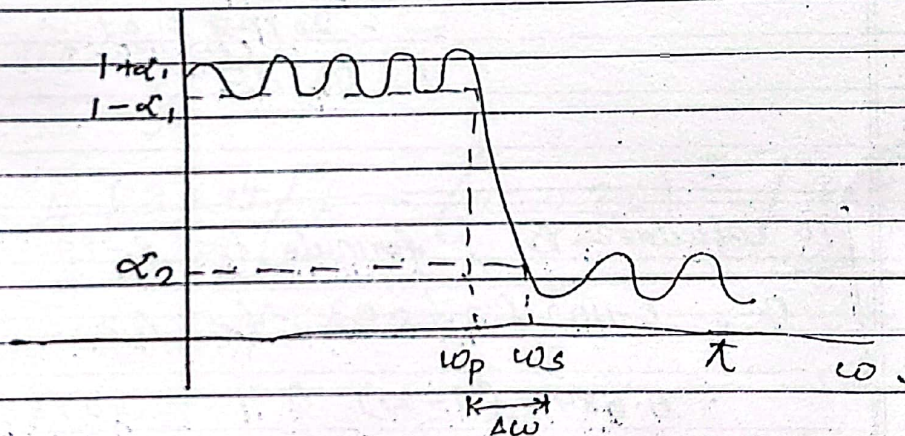
Design a linear FIR phase filter using Kaiser window to meet following specs.

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \quad 0 \leq |\omega| \leq 0.19\pi$$

$$|H(e^{j\omega})| \leq 0.01, \quad 0.21\pi \leq |\omega| \leq \pi$$

⇒ Solution:

Here;



From figure & given sp:

$$\omega_p = 0.19\pi \quad \alpha_2 = 0.01$$

$$\omega_s = 0.21\pi \quad \alpha_1 = 0.01$$

$$1+\alpha_1 = 1.01$$

$$\alpha_1 = 0.01$$

$$1-\alpha_1 = 0.99$$

$$\alpha_1 = 0.01$$

$$\text{Transition width } (\Delta\omega) = \omega_s - \omega_p$$

$$= 0.21\pi - 0.19\pi$$

$$= 0.02\pi$$

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

$$= \frac{0.19\pi + 0.21\pi}{2}$$

$$= \frac{0.40\pi}{2}$$

$$= 0.20\pi$$

Minimum Ripple (α) = 0.01

$$\begin{aligned} \text{Attenuation (A)} &= -20 \log_{10} \alpha \\ &= -20 \log_{10} 0.01 \\ &= 40 \text{ dB} \end{aligned}$$

To calculate B, formula for β is.

$$\beta = 0.1102 (A - 87) \text{ for } A > 50$$

$$= 0.5842 (A - 21) 0.4 + 0.07886 (A - 21) \text{ for } 21 \leq A \leq 50$$

$$= 0 \text{ for } A < 21,$$

Since our A is 40 dB, we choose ^{2nd} formula.

$$\therefore \beta = 3.395$$

$$M = \frac{A-8}{2.285 \Delta \omega}$$

$$= 222.88$$

$$M \approx 224$$

In case of all filter make M as odd, but in case of this filter, make even.

Now,

$$\alpha = \frac{M}{2}$$

$$= \frac{224}{2}$$

$$\alpha = 112$$

$$\omega(n) = J_0 \left\{ 3.895 \left[1 - \left(\frac{n-112}{112} \right)^2 \right]^{1/2} \right\}$$

$$J_0(3.895)$$

J_0 = Bessel's function.

$$J_0 x = 1 - \frac{0.25}{x} + \dots$$

Now,

$$H_d(\omega) = e^{-j\omega\tau}, \quad -\omega_c \leq |\omega| \leq \omega_c$$

$$= 0, \quad \text{otherwise}$$

$$h_d(n) = \frac{\sin \omega_c (n - \tau)}{\pi (n - \tau)}, \quad n \neq \tau$$

$$= \frac{\omega_c}{\pi}, \quad n = \tau$$

$$h_d(n) = \frac{\sin 0.2 (n - 112)}{\pi (n - 112)}, \quad n \neq 112$$

$$= \frac{0.2}{\pi}, \quad n = 112$$

$n = 0 \dots M$
 $h = 0 \dots 102$